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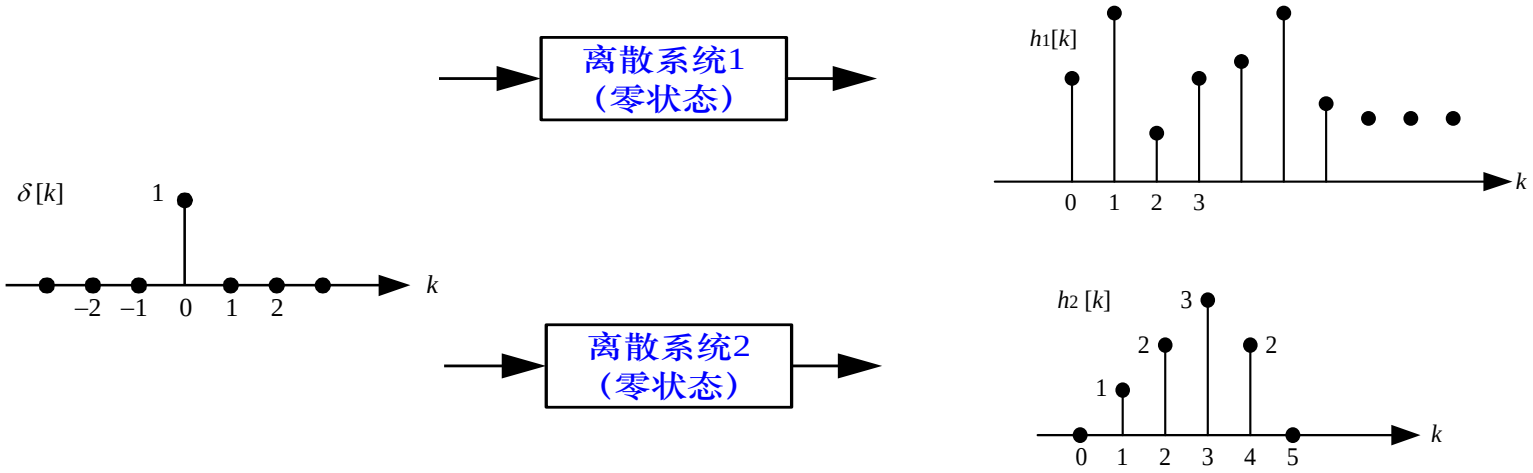
The unit impulse response represents the characteristics of the system

- The unit impulse response of the cascading system
- Unit impulse response of parallel systems
- Unit impulse response and system causality
- Unit impulse response and system stability

The unit impulse response represents the characteristics of the system



The unit impulse response $h[k]$ is only related to the system itself. $h[k]$ is used to represent the time domain characteristics of discrete LTI systems.

The unit impulse response represents the characteristics of the system

The unit impulse response of several common systems

Distortion-free transmission system

input: $d[k] \Rightarrow h[k] = K \cdot \delta[k - k_{\mathrm{d}}]$

K is the normal number, and k_d is the delay time after the input signal passes through the system.

Summator

input: $d[k] \Rightarrow h[k] = u[k]$

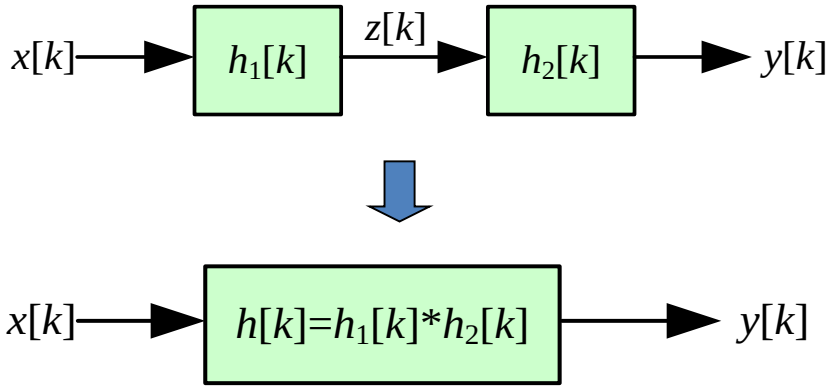
Diff

input: $d[k] \Rightarrow h[k] = \delta[k] - \delta[k-1]$

Unit delayer

input: $d[k] \Rightarrow h[k] = \delta[k-1]$

1. Unit impulse response of cascaded systems

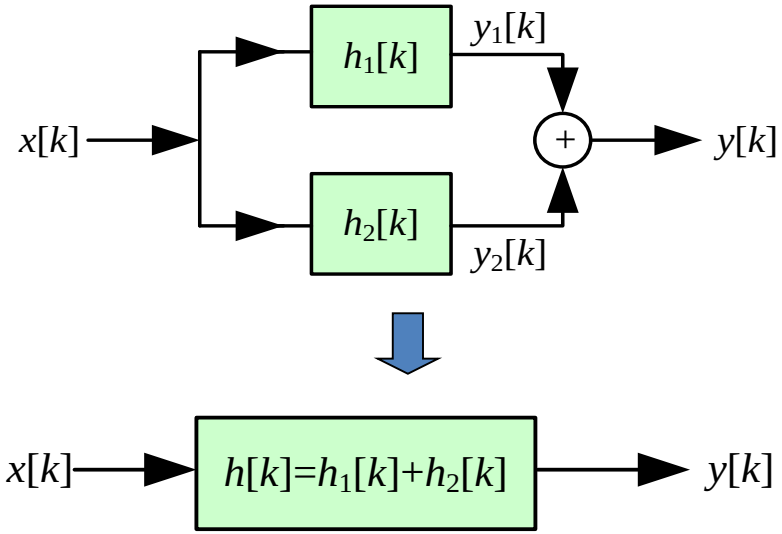


$$z[k] = x[k] * h_1[k]$$

$$y[k] = z[k] * h_2[k]$$

$$y[k] = x[k] * \{\underbrace{h_1[k] * h_2[k]}_{h[k]}\}$$

2. Unit impulse response of parallel systems

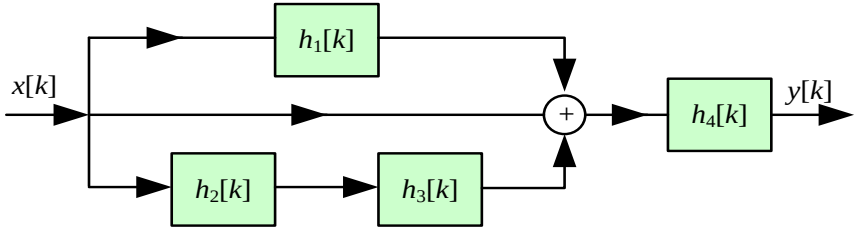


$$y_1[k] = x[k] * h_1[k]$$

$$y_2[k] = x[k] * h_2[k]$$

$$y[k] = x[k] * \{\underbrace{h_1[k] + h_2[k]}_{h[k]}\}$$

[Example] Finds the unit impulse response of the graphical system, wherein $h_1[k]$ $=2^k u[k]$, $h_2[k] = \delta[k-1]$, $h_3[k] = 3^k u[k]$, $h_4[k] = u[k]$



Subsystem $h_2[k]$ is cascaded with $h_3[k]$, $h_1[k]$ branch, all-through branch and $h_2[k]$ $h_3[k]$ cascade branch in parallel, and then cascaded with $h_4[k]$.

All-through branch satisfied $y[k] = x[k] * h[k] = x[k]$

The unit impulse response of the all-through discrete system is $\delta[k]$

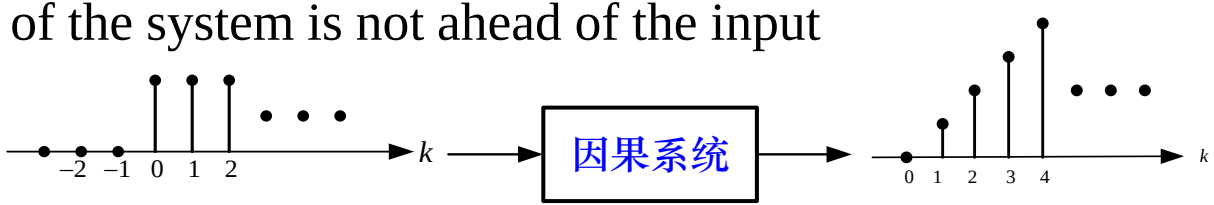
$$h[k] = \{h_1[k] + \delta[k] + h_2[k] * h_3[k]\} * h_4[k]$$

$$= 2(2)^k u[k] + [1.5(3)^{k-1} - 0.5]u[k-1]$$

3. Unit impulse response and system causality

Causal System Definition:

If the output of the k_0 moment of the system is only related to the k_0 moment and the previous input signal, the system is called a causal system. In short, the output of the system is not ahead of the input



Discrete LTI systems are sufficiently necessary for causal systems

$$h[k] = 0, \quad k < 0$$

3. Unit impulse response and system causality

[Example] Determine whether the following discrete LTI system is a causal system.

$$h[k]=\cos[\frac{\pi}{2}k]u[k]$$

解: The system is a discrete LTI system

When $k<0$时，$u[k]=0$

So satisfied $h[k]=0$, $k<0$。

This system is a causal system.

4. Unit impulse response and system stability

Stable System Definition:

If the system is bounded to any bounded input, its output is also bounded. The system is called a stable system (BIBO stable)

A discrete LTI system is a sufficient requirement for stabilizing the system

$$\sum_{k=-\infty}^{\infty} \left|h[k]\right| = S < \infty$$

4. Unit impulse response and system stability

[Example] Determine whether the following discrete LTI system is a stable system.

$$h[k] = 2^k \{u[k] - u[k-3]\}$$

Solution: The charging condition for the stability of a discrete LTI system is

$$\sum_{k=-\infty}^{\infty} |h[k]| = S < \infty$$

$$\sum_{k=-\infty}^{\infty} |h[k]| = \sum_{k=0}^{2} |2^k| = 1 + 2 + 4 = 7 < \infty$$

To meet the charging conditions of stability, the system is a stable system.

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!